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PHYS 3316 — Quantum Mechanics I

Notebook 4: Localized Potentials — The Delta Function and Finite Square Well

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Overview

The infinite square well introduced the idea that boundary conditions lead naturally to quantized energy levels. In this notebook, we consider two more realistic one-dimensional potentials: the attractive delta-function potential and the finite square well.

Unlike the infinite well, these systems allow the wavefunction to extend into classically forbidden regions, introducing the important quantum phenomenon of **tunneling**. Although both models are idealized, they capture many of the essential features of bound states in atoms, molecules, semiconductor quantum wells, and nanoscale devices.

The attractive delta-function potential provides the simplest example of a localized potential that supports a bound state. Despite its simplicity, it illustrates how an arbitrarily weak attractive potential can bind a particle in one dimension.

The finite square well extends these ideas by introducing a potential of finite width and depth. In this case, the Schrödinger equation must be solved in multiple regions and the solutions matched at the boundaries. The resulting transcendental equations generally require numerical methods to determine the allowed bound-state energies.

Throughout this notebook, analytical derivations are combined with computational methods to investigate the properties of bound states and tunneling.

Learning Objectives

After completing this notebook, you should be able to

- derive the bound-state solution for the attractive delta-function potential,

- apply continuity and derivative boundary conditions to piecewise potentials,
 - derive the transcendental equations for the finite square well,
 - solve transcendental equations numerically using Python,
 - distinguish between even- and odd-parity eigenstates,
 - visualize wavefunction penetration into classically forbidden regions,
 - investigate how bound-state energies depend on the well depth and width,
 - recover the infinite square well as a limiting case of the finite well.
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Physical Models

Attractive Delta-Function Potential

The attractive delta-function potential is

$$V(x) = -\alpha \delta(x), \quad \alpha > 0,$$

where $\delta(x)$ is the Dirac delta function.

Although infinitely narrow, the potential supports exactly one bound state.

Finite Square Well

The finite square well is defined by

$$V(x) = \begin{cases} -V_0, & |x| < a, \\ 0, & |x| \geq a, \end{cases}$$

where

- V_0 is the well depth,
- $2a$ is the well width.

Unlike the infinite well, the wavefunction extends beyond the boundaries of the well and decays exponentially outside the classically allowed region.

Key Concepts

This notebook introduces several fundamental ideas that appear throughout quantum mechanics.

Localized Bound States

A particle is said to be bound when its energy is less than the potential outside the well. Bound-state wavefunctions are normalizable and decay exponentially away from the potential.

Quantum Tunneling

Unlike a classical particle, a quantum particle has a nonzero probability of being found in a region where

$$E < V(x).$$

This exponential penetration into forbidden regions is known as **tunneling** and is one of the defining features of quantum mechanics.

Boundary Conditions

Solutions to the Schrödinger equation must satisfy the appropriate boundary conditions.

For the finite square well,

- the wavefunction is continuous,
- its first derivative is continuous.

For the delta-function potential,

- the wavefunction remains continuous,
 - its derivative has a finite discontinuity determined by the strength of the delta function.
-

Numerical Root Finding

The allowed energies of the finite square well satisfy transcendental equations that generally cannot be solved analytically.

Instead, numerical methods such as

```
scipy.optimize.root_scalar()  
or
```

```
scipy.optimize.brentq()  
are used to determine the bound-state energies.
```

Computational Workflow

Throughout this notebook you will

1. derive analytical expressions for bound states,
 2. solve transcendental equations numerically,
 3. compute normalized wavefunctions,
 4. visualize tunneling and exponential decay,
 5. compare finite and infinite square wells,
 6. investigate how bound states evolve as potential parameters change.
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Python Packages

```
import numpy as np
import matplotlib.pyplot as plt

from scipy.integrate import simpson
from scipy.optimize import root_scalar
```

Suggested Numerical Parameters

Unless otherwise specified,

```
# Atomic units
hbar = 1.0
m = 1.0

# Spatial grid
xmin = -10.0
xmax = 10.0
Nx = 2000

# Finite square well
V0 = 5.0
a = 1.0
```

These values may be modified throughout the notebook to explore different physical regimes.

Computational Themes

This notebook emphasizes several numerical techniques commonly used in computational quantum mechanics:

- Numerical root finding
- Piecewise wavefunction construction
- Numerical normalization

- Parameter sweeps
- Visualization of bound states
- Comparison of analytical and numerical solutions

Many quantum systems do not possess exact analytical solutions. The methods introduced here provide a foundation for solving more realistic quantum problems encountered later in the course.

Problem 1 — Bound State of the Attractive Delta-Function Potential

Consider a particle of mass m moving in the one-dimensional potential

$$V(x) = -\alpha \delta(x), \quad \alpha > 0.$$

This potential is zero everywhere except at $x = 0$, where it is infinitely narrow and attractive.

Despite its extreme simplicity, the attractive delta-function potential supports a bound state.

Physical Objective

Determine the bound-state energy and normalized wavefunction of the attractive delta-function potential.

This problem introduces an important boundary condition: the wavefunction remains continuous, but its derivative is discontinuous at the location of the delta function.

Mathematical Background

The time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} - \alpha \delta(x)\psi(x) = E\psi(x).$$

For a bound state,

$$E < 0.$$

It is convenient to write

$$E = -\frac{\hbar^2 \kappa^2}{2m}, \quad \kappa > 0.$$

For $x \neq 0$, the potential vanishes, so the Schrödinger equation becomes

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi(x).$$

Substituting the bound-state energy gives

$$\frac{d^2\psi}{dx^2} = \kappa^2\psi.$$

Analytical Tasks

(a) Solve the Schrödinger equation away from the origin

Find the general solution in the two regions

$$x < 0$$

and

$$x > 0.$$

Use the requirement that a bound-state wavefunction must remain finite and normalizable as

$$x \rightarrow \pm\infty$$

to show that

$$\psi(x) = \begin{cases} Ae^{\kappa x}, & x < 0, \\ Be^{-\kappa x}, & x > 0. \end{cases}$$

(b) Apply continuity of the wavefunction

The wavefunction must be continuous at the origin:

$$\psi(0^-) = \psi(0^+).$$

Use this condition to show that

$$A = B.$$

Hence write the bound-state wavefunction as

$$\psi(x) = Ae^{-\kappa|x|}.$$

(c) Derive the derivative discontinuity

Integrate the Schrödinger equation over the interval

$$-\epsilon < x < \epsilon$$

and then take the limit

$$\epsilon \rightarrow 0.$$

Show that

$$\psi'(0^+) - \psi'(0^-) = -\frac{2m\alpha}{\hbar^2}\psi(0).$$

This jump condition replaces the usual derivative continuity encountered for finite potentials.

SOLUTION:

Problem 2 — Exploring the Delta-Function Bound State Numerically

In Problem 1, you derived the normalized bound-state wavefunction and energy for the attractive delta-function potential

$$V(x) = -\alpha \delta(x), \quad \alpha > 0.$$

The bound-state solution is

$$\psi(x) = \sqrt{\kappa} e^{-\kappa|x|},$$

where

$$\kappa = \frac{m\alpha}{\hbar^2},$$

and

$$E = -\frac{\hbar^2 \kappa^2}{2m} = -\frac{m\alpha^2}{2\hbar^2}.$$

In this problem, you will use Python to visualize the bound state and investigate how its energy and spatial extent depend on the strength of the attractive potential.

Physical Objective

Explore the relationship between

- the strength of the attractive potential,
- the bound-state energy,
- the localization length,
- the shape of the probability density.

This problem also provides practice with numerical normalization and parameter sweeps.

Computational Setup

Use atomic units,

$$\hbar = m = 1.$$

In these units,

$$\kappa = \alpha,$$

$$E = -\frac{\alpha^2}{2},$$

and

$$\psi(x) = \sqrt{\alpha} e^{-\alpha|x|}.$$

Begin with

```
import numpy as np
import matplotlib.pyplot as plt
```

```
from scipy.integrate import simpson
```

Use the following initial parameters:

```
hbar = 1.0
```

```
m = 1.0
```

```
alpha = 1.0
```

```
xmax = 10.0
```

```
Nx = 2001
```

```
x = np.linspace(-xmax, xmax, Nx)
```

(a) Define the Bound-State Quantities

Write Python functions that return

- the inverse decay length κ ,
- the bound-state energy E ,
- the normalized wavefunction $\psi(x)$.

For example,

```
def kappa(alpha, m=1.0, hbar=1.0):
    # Your code here
    pass

def bound_state_energy(alpha, m=1.0, hbar=1.0):
    # Your code here
    pass

def psi_delta(x, alpha, m=1.0, hbar=1.0):
    # Your code here
    pass
```

Test your functions using

`alpha = 1.0`
and print the values of κ and E .

(b) Plot the Wavefunction and Probability Density

Calculate

$$\psi(x)$$

and

$$|\psi(x)|^2$$

on the spatial grid.

Create separate plots of

1. the bound-state wavefunction,
2. the probability density.

Label the axes clearly.

Because the delta function cannot be plotted as an ordinary function, indicate the location of the potential with a vertical line at

$$x = 0.$$

(c) Verify Normalization Numerically

Evaluate

$$\int_{-x_{\max}}^{x_{\max}} |\psi(x)|^2 dx$$

using numerical integration.

For example,

```
normalization = simpson(np.abs(psi)**2, x=x)
```

Calculate the normalization error,

$$\epsilon_{\text{norm}} = \left| 1 - \int |\psi(x)|^2 dx \right|.$$

Print both the numerical normalization and the error.

Explain why the result may differ slightly from exactly one.

(d) Investigate the Spatial Domain

Repeat the normalization calculation for several values of $x_{\max}$:

```
xmax_values = [2.0, 4.0, 6.0, 8.0, 10.0]
```

Keep the grid spacing sufficiently small for each calculation.

Construct a table containing

$x_{\max}$	Numerical normalization	Normalization error
------------	-------------------------	---------------------

Plot the normalization error as a function of $x_{\max}$.

Use a logarithmic vertical scale if useful.

(e) Vary the Potential Strength

Repeat the calculation for

```
alpha_values = [0.25, 0.5, 1.0, 2.0]
```

For each value of α , calculate

$$\kappa = \alpha,$$

$$E = -\frac{\alpha^2}{2},$$

and the characteristic localization length

$$\ell = \frac{1}{\kappa}.$$

Construct a table containing

$$\alpha \quad \kappa \quad E \quad \ell$$

(f) Compare the Probability Densities

Plot

$$|\psi(x)|^2$$

for all values of α on the same axes.

Use a legend to identify each curve.

Describe how the probability density changes as α increases.

(g) Verify the Energy Scaling

The analytical result predicts

$$E \propto -\alpha^2.$$

Calculate the energy for a larger set of values, such as

```
alpha_scan = np.linspace(0.1, 2.0, 100)
```

Plot E as a function of α .

On the same figure, compare the numerical values with the analytical curve

$$E = -\frac{\alpha^2}{2}.$$

(h) Verify the Localization Scaling

The localization length is

$$\ell = \frac{1}{\alpha}.$$

Plot ℓ as a function of α .

Explain why a stronger attractive potential produces a more localized bound state.

Expected Output

Your notebook should include

- Python functions for κ , E , and $\psi(x)$,
 - a plot of the bound-state wavefunction,
 - a plot of the probability density,
 - numerical verification of normalization,
 - a convergence study with respect to $x_{\max}$,
 - a table of α , κ , E , and ℓ ,
 - a comparison of probability densities for several values of α ,
 - plots showing the energy and localization-length scaling.
-

Physical Interpretation

Answer the following questions.

1. What happens to the energy as the attraction strength α increases?
 2. Why does the energy become more negative for a stronger potential?
 3. What happens to the spatial width of the probability density as α increases?
 4. Why is a larger spatial domain required when α is small?
 5. How are the localization length and bound-state energy related?
 6. Does the bound state disappear as α approaches zero?
 7. What happens to the wavefunction in the limit $\alpha \rightarrow 0^+$?
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Optional Challenge — Probability Outside a Chosen Region

Calculate the probability of finding the particle outside the interval

$$|x| < L.$$

That is, evaluate

$$P_{\text{outside}} = \int_{-\infty}^{-L} |\psi(x)|^2 dx + \int_L^{\infty} |\psi(x)|^2 dx.$$

Derive an analytical expression for P_{outside} and verify it numerically.

Investigate how this probability depends on

- L ,
- α .

Explain how this quantity provides a direct measure of the spatial extent of the bound state.

SOLUTION:

Problem 3 — Bound States of the Finite Square Well

Consider the symmetric finite square well

$$V(x) = \begin{cases} -V_0, & |x| < a, \\ 0, & |x| \geq a, \end{cases}$$

where $V_0 > 0$ is the well depth and $2a$ is the total width.

Unlike the infinite square well, the potential outside the well is finite. Therefore, a bound-state wavefunction does not vanish at $x = \pm a$. Instead, it extends into the classically forbidden region and decays exponentially.

Physical Objective

Derive the equations that determine the allowed bound-state energies of the finite square well.

This problem emphasizes

- piecewise solutions of the Schrödinger equation,
- parity,
- boundary matching,
- exponential penetration into classically forbidden regions.

The energy equations obtained here are transcendental and will be solved numerically in Problem 4.

Mathematical Background

For a bound state,

$$-V_0 < E < 0.$$

Inside the well, define

$$k = \frac{\sqrt{2m(E + V_0)}}{\hbar}.$$

Outside the well, define

$$\kappa = \frac{\sqrt{-2mE}}{\hbar}.$$

Because

$$E + V_0 > 0,$$

the wavefunction is oscillatory inside the well.

Because

$$E < 0,$$

the wavefunction decays exponentially outside the well.

The two wave numbers satisfy

$$k^2 + \kappa^2 = \frac{2mV_0}{\hbar^2}.$$

(a) Solve the Schrödinger Equation in Each Region

Divide space into three regions:

- Region I: $x < -a$
- Region II: $-a < x < a$
- Region III: $x > a$

Write the general solution in each region.

Use normalizability to eliminate any exponential that grows as $x \rightarrow \pm\infty$.

Show that the wavefunction may be written as

$$\psi(x) = \begin{cases} Ae^{\kappa x}, & x < -a, \\ B \cos(kx) + C \sin(kx), & -a < x < a, \\ De^{-\kappa x}, & x > a. \end{cases}$$

(b) Use the Symmetry of the Potential

The finite square well is symmetric:

$$V(-x) = V(x).$$

Therefore, the bound-state eigenfunctions may be chosen to have definite parity.

Even-parity states satisfy

$$\psi(-x) = \psi(x),$$

while odd-parity states satisfy

$$\psi(-x) = -\psi(x).$$

Explain why it is sufficient to solve the problem for $x \geq 0$ and then extend the result to $x < 0$ using parity.

Even-Parity Bound States

For an even state, use

$$\psi_{\text{even}}(x) = \begin{cases} A \cos(kx), & 0 \leq x < a, \\ Be^{-\kappa x}, & x \geq a. \end{cases}$$

(c) Match the Even Wavefunction

Continuity of the wavefunction at $x = a$ requires

$$A \cos(ka) = Be^{-\kappa a}.$$

Continuity of the derivative requires

$$-kA \sin(ka) = -\kappa Be^{-\kappa a}.$$

Divide the derivative condition by the wavefunction condition and show that

$$\boxed{k \tan(ka) = \kappa}.$$

This is the transcendental equation for the even-parity bound states.

Odd-Parity Bound States

For an odd state, use

$$\psi_{\text{odd}}(x) = \begin{cases} A \sin(kx), & 0 \leq x < a, \\ Be^{-\kappa x}, & x \geq a. \end{cases}$$

The solution for $x < 0$ is obtained from odd parity.

(d) Match the Odd Wavefunction

Continuity of the wavefunction at $x = a$ gives

$$A \sin(ka) = Be^{-\kappa a}.$$

Continuity of the derivative gives

$$kA \cos(ka) = -\kappa Be^{-\kappa a}.$$

Divide the derivative condition by the wavefunction condition and show that

$$\boxed{-k \cot(ka) = \kappa}.$$

This is the transcendental equation for the odd-parity bound states.

(e) Introduce Dimensionless Variables

Define

$$z = ka$$

and

$$z_0 = \frac{a\sqrt{2mV_0}}{\hbar}.$$

Using

$$k^2 + \kappa^2 = \frac{2mV_0}{\hbar^2},$$

show that

$$\kappa a = \sqrt{z_0^2 - z^2}.$$

The even-state equation becomes

$$z \tan z = \sqrt{z_0^2 - z^2}.$$

The odd-state equation becomes

$$-z \cot z = \sqrt{z_0^2 - z^2}.$$

The allowed bound states satisfy

$$0 < z < z_0.$$

(f) Relate z to the Energy

Starting from

$$z = \kappa a,$$

show that

$$E = -V_0 + \frac{\hbar^2 z^2}{2ma^2}.$$

Equivalently, using

$$z_0^2 = \frac{2mV_0a^2}{\hbar^2},$$

show that

$$E = V_0 \left(\frac{z^2}{z_0^2} - 1 \right).$$

Explain why all bound-state roots must satisfy

$$0 < z < z_0.$$

(g) Sketch the Transcendental Equations

For the even states, sketch or plot the two functions

$$y = z \tan z$$

and

$$y = \sqrt{z_0^2 - z^2}.$$

For the odd states, sketch or plot

$$y = -z \cot z$$

and

$$y = \sqrt{z_0^2 - z^2}.$$

Each intersection corresponds to an allowed bound-state energy.

Use a representative value such as

$$z_0 = 4.$$

Do not attempt to solve the equations numerically yet.

Expected Output

Your work should include

- the piecewise form of the bound-state wavefunction,
 - the even-parity transcendental equation,
 - the odd-parity transcendental equation,
 - the dimensionless equations,
 - the relation between each root z and the energy,
 - sketches or plots showing how the bound-state roots arise.
-

Physical Interpretation

Answer the following questions.

1. Why must a bound-state energy satisfy $-V_0 < E < 0$?
2. Why is the wavefunction oscillatory inside the well?
3. Why is the wavefunction exponentially decaying outside the well?
4. Why are both ψ and ψ' continuous at $x = \pm a$?
5. Why can the eigenstates be classified as even or odd?
6. Why does the finite well support only a finite number of bound states?
7. What is the physical meaning of the decay constant κ ?

8. What happens to the exponential tails as the energy approaches zero from below?
 9. How do the finite-well boundary conditions differ from those of the infinite square well?
-

Connection to the Infinite Square Well

In the limit

$$V_0 \rightarrow \infty,$$

the decay constant outside the well becomes very large, and the wavefunction is confined more strongly to the region

$$-a < x < a.$$

Explain qualitatively why the finite-well energy levels should approach those of an infinite square well of width $2a$.

Optional Challenge — Bound-State Counting

Use the dimensionless parameter

$$z_0 = \frac{a\sqrt{2mV_0}}{\hbar}$$

to estimate how the number of bound states changes as either

- the well depth V_0 increases,
- the half-width a increases.

Explain why a wider or deeper well can support more bound states.

SOLUTION:

Problem 4 — Numerical Bound States of the Finite Square Well

In Problem 3, you derived the dimensionless bound-state equations for the symmetric finite square well:

$$z \tan z = \sqrt{z_0^2 - z^2}$$

for even-parity states, and

$$-z \cot z = \sqrt{z_0^2 - z^2}$$

for odd-parity states.

Here,

$$z = ka$$

and

$$z_0 = \frac{a\sqrt{2mV_0}}{\hbar}.$$

Each valid root in the interval

$$0 < z < z_0$$

corresponds to one bound-state energy.

Physical Objective

Determine the bound-state energies of the finite square well numerically.

This problem introduces numerical root-finding for transcendental equations and shows how parity organizes the bound-state spectrum.

Computational Setup

Use atomic units,

$$\hbar = m = 1,$$

and begin with

```
import numpy as np
import matplotlib.pyplot as plt

from scipy.optimize import root_scalar
Choose
```

```
a = 1.0
V0 = 8.0
```

Then calculate

`z0 = a * np.sqrt(2.0 * V0)`

(a) Define the Transcendental Functions

Define

$$f_{\text{even}}(z) = z \tan z - \sqrt{z_0^2 - z^2},$$

and

$$f_{\text{odd}}(z) = -z \cot z - \sqrt{z_0^2 - z^2}.$$

Write Python functions for both equations.

For example,

```
def f_even(z, z0):
    # Your code here
    pass
```

```
def f_odd(z, z0):
    # Your code here
    pass
```

Recall that

$$\cot z = \frac{1}{\tan z}.$$

(b) Plot the Root Equations

Plot the following functions over

$$0 < z < z_0 :$$

$$y_{\text{even}}(z) = z \tan z,$$

$$y_{\text{odd}}(z) = -z \cot z,$$

and

$$y_{\text{rhs}}(z) = \sqrt{z_0^2 - z^2}.$$

Because the tangent and cotangent functions contain vertical asymptotes, limit the vertical plotting range to make the intersections visible.

For example,

```
plt.ylim(0, z0 + 1)
```

Identify approximately where the even and odd roots occur.

(c) Identify the Valid Root Intervals

The tangent and cotangent functions are discontinuous at specific values of z .

Even roots occur in intervals where

$$z \tan z > 0.$$

Odd roots occur in intervals where

$$-z \cot z > 0.$$

For the first few branches, the relevant intervals are

Even states

$$0 < z < \frac{\pi}{2},$$

$$\pi < z < \frac{3\pi}{2},$$

$$2\pi < z < \frac{5\pi}{2},$$

and so forth.

Odd states

$$\frac{\pi}{2} < z < \pi,$$

$$\frac{3\pi}{2} < z < 2\pi,$$

$$\frac{5\pi}{2} < z < 3\pi,$$

and so forth.

Retain only those portions lying below z_0 .

(d) Find the Roots Numerically

Use a bracketing root-finding method such as

```
root_scalar()
```

with

method="brentq"
to locate each root.

For example,

```
solution = root_scalar(
    f_even,
    args=(z0,),
    bracket=(z_left, z_right),
    method="brentq"
)
```

Avoid evaluating the function exactly at a tangent or cotangent singularity. A small offset may be used:

eps = 1.0e-6

Find all valid even and odd roots in the interval

$$0 < z < z_0.$$

(e) Convert the Roots to Energies

For each root, calculate the bound-state energy using

$$E = -V_0 + \frac{\hbar^2 z^2}{2ma^2}.$$

In atomic units,

$$E = -V_0 + \frac{z^2}{2a^2}.$$

Alternatively,

$$E = V_0 \left(\frac{z^2}{z_0^2} - 1 \right).$$

Verify that each energy satisfies

$$-V_0 < E < 0.$$

(f) Construct the Bound-State Table

Sort all states in order of increasing energy.

Construct a table containing

State	Parity	Root z	Energy E
0			
1			
2			

Check that the parity alternates as the energy increases.

(g) Compare with the Infinite Square Well

For an infinite square well extending from $-a$ to a , the energies are

$$E_n^{(\infty)} = \frac{n^2 \pi^2 \hbar^2}{8ma^2}, \quad n = 1, 2, 3, \dots$$

The finite-well energies are measured relative to the zero of potential outside the well. To compare the spectra inside the well, define

$$\mathcal{E} = E + V_0.$$

For each finite-well state, calculate

$$\mathcal{E} = \frac{\hbar^2 z^2}{2ma^2}.$$

Compare these values with the corresponding infinite-well energies.

Are the finite-well energies above or below the infinite-well values?

Explain your result physically.

(h) Vary the Well Depth

Repeat the root-finding calculation for several values of V_0 , such as

`V0_values = [1.0, 3.0, 5.0, 8.0, 12.0]`

For each value, determine

- z_0 ,
- the number of bound states,
- the energy of each bound state.

Construct a table containing

V_0	z_0	Number of bound states
-------	-------	------------------------

(i) Plot the Bound-State Energies

Plot the bound-state energies as horizontal lines over the finite square-well potential.

Your figure should show

- the potential minimum at $-V_0$,
- the continuum threshold at $E = 0$,
- each bound-state energy,
- the parity of each state.

A schematic potential may be generated with

```
V = np.where(np.abs(x) < a, -V0, 0.0)
```

Expected Output

Your notebook should include

- plots of the transcendental equations,
- all valid even and odd roots,
- a table of roots and bound-state energies,
- a comparison with the infinite square well,
- a table showing the number of bound states as V_0 changes,
- a potential-energy diagram containing the bound-state levels.

Physical Interpretation

Answer the following questions.

1. Why must the root search be divided into separate intervals?
2. Why should a root-finding algorithm not cross a tangent or cotangent asymptote?
3. Why does parity alternate between successive bound states?
4. Why does a deeper well support more bound states?
5. Why is the number of bound states finite?
6. Why are the finite-well energies lower than the corresponding infinite-well energies when measured from the bottom of the well?
7. What happens to the highest bound-state energy as a new state first appears?

8. Why does the ground state exist even for a very shallow attractive well?

Numerical Note

A sign change alone does not always imply a physical root if the interval contains a singularity. Before calling the root solver, confirm that

- both endpoints lie on the same continuous branch,
- the function is finite at both endpoints,
- the function changes sign across the interval.

This distinction is essential when solving transcendental equations numerically.

Optional Challenge — Automatic Root Search

Write a function

```
def finite_well_roots(V0, a=1.0, m=1.0, hbar=1.0):  
    ...
```

that automatically

1. calculates z_0 ,
2. constructs every valid even and odd interval,
3. avoids the singular endpoints,
4. finds all bound-state roots,
5. calculates the corresponding energies,
6. returns the states sorted by energy.

Test the function over a range of well depths.